CSE1905- Technical Answers for Real World Problems (TARP)

MAINTAINIQ

An Explainable MongoDB-Native Decision Support Platform for Industrial Asset Health Monitoring



BASE PAPER DETAILS

Base Paper: "An Autoencoder-Based Framework for Time-Series Anomaly Detection in Metro Train Air Production Units"

Key Findings:

- Autoencoder achieved **F1 > 0.96** and **AUC > 0.96**
- Detected anomalies up to **2 hours before failure**
- **Research Gap Identified:** "The development of a decision-support management system is crucial for determining when and how to trigger alarms for maintenance personnel"

Base Paper	MaintainIQ
Detects anomalies	Explains why asset is flagged
Binary output	0-100 health score (DAHI)
No prioritization	Ranked maintenance queue (MAPE)
No decision support	Rule-based recommendations (EME)
No NL interface	Natural language queries (AMA)
No NoSQL focus	MongoDB-native platform

PROBLEM STATEMENT & OBJECTIVES

Problem Statement: Industrial facilities generate massive volumes of continuous sensor data from machinery, yet maintenance teams struggle to convert this raw data into timely, understandable, and actionable decisions.

Objectives

1. Leverage MongoDB as computational backbone (not just storage)
2. Compute interpretable health scores (0-100) via aggregation pipelines
3. Prioritize maintenance by business impact
4. Provide explainable recommendations with actions
5. Enable natural-language querying via agentic assistant

Issue	Impact
Alert Fatigue	Engineers can't prioritize hundreds of alerts
Black-Box AI	"91% failure" with no explanation → no trust
Reactive Maintenance	Fix after breakdown → costly downtime
Lost Knowledge	Senior engineers retire → experience lost
Inflexible Databases	Adding sensors requires ALTER TABLE

HARDWARE & SOFTWARE CONFIGURATION

Hardware Requirements

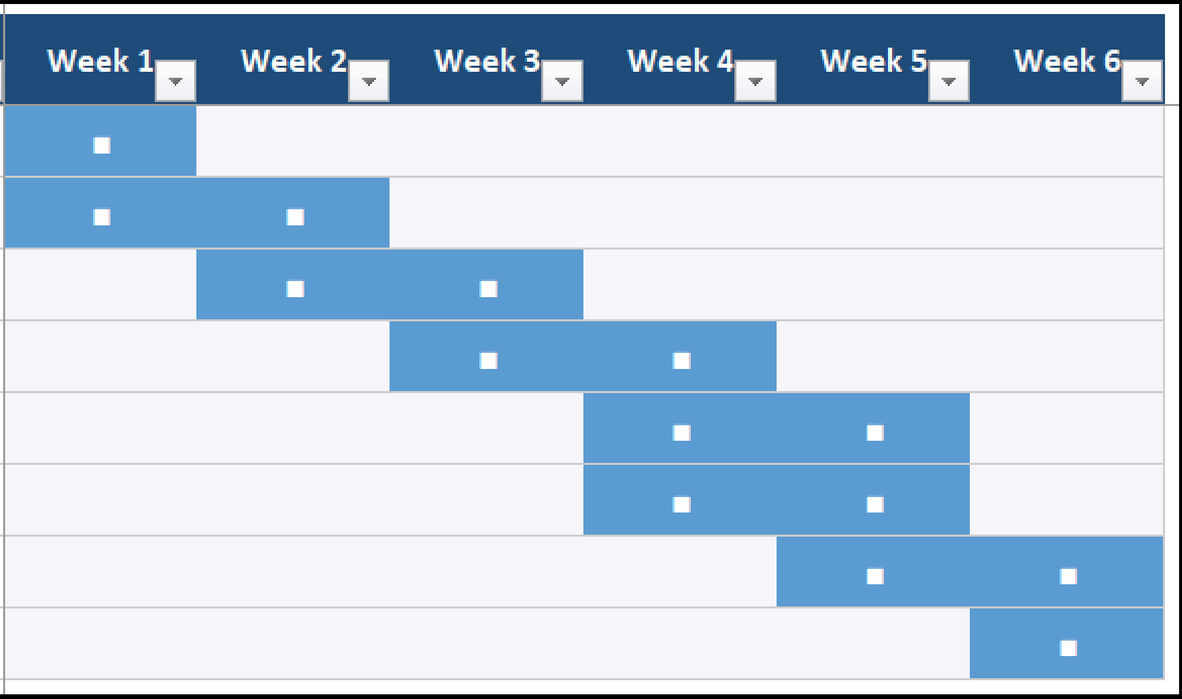
Component	Specification
Development Machine	16GB+ RAM, 4+ cores, 256GB+ SSD
Database Server	8GB+ RAM, 50GB+ storage

Software Stack

Layer	Technology	Version	Purpose
Database	MongoDB	7.0+	Time-Series + Aggregation
Database (Benchmark)	MySQL	8.0+	Relational comparison
Model	TensorFlow/Keras	2.13+	LSTM Autoencoder
Backend	Python	3.9+	Data processing + API
Dashboard	Streamlit	1.25+	Real-time visualization
NLP	LangChain + LLM API	Latest	Natural language

GANTT CHART

Task	Start Week	End Week	Duration
Environment Setup	1	1	1
Data Import to MongoDB	1	2	2
LSTM Autoencoder Training	2	3	2
DAHI Health Score Implementation	3	4	2
MAPE Prioritization Engine	4	5	2
EME Explanation Engine	4	5	2
Dashboard + AMA Agent	5	6	2
Benchmarking & Documentation	6	6	1



LITERATURE REVIEW: 21 PAPERS

Key Literature Insights

The literature review of 21 papers reveals critical findings that justify our approach:

S.No	Title & Year	Dataset	Algorithm & Accuracy	Limitation
1	A predictive maintenance model for Internet of Things devices using long short-term memory and one-dimensional dilated group convolution with residual connection (2024)	FD001 (C-MAPSS); Real-world industrial dataset (11362 samples, 82 features)	LSTM + 1DDGCR; FD001: 0.867 (train), 0.8355 (test); Real: 0.893; 3-class: 0.9539	Performance drops with more class labels; Hyperparameter dependent via BOA
2	A survey of deep learning-driven architecture for predictive maintenance (2024)	106 papers (2017-2023); C-MAPSS; Various industrial datasets	Survey: DNN, CNN, SAE, DBN, DRNN; DNN >99%; DRNN best for time-series; CNN 92.7%	Lack of standardization; Dataset accessibility; Interpretability challenges
3	A Systematic Review of Digital Twin-Driven Predictive Maintenance in Industrial Engineering: Taxonomy, Architectural Elements, and Future Research Directions (2025)	62 papers; Marine, Auto, Energy, Aviation, Railway, Manufacturing	PRISMA SLR; Case Study: AUC 0.99, Accuracy 95%, F1 >0.90; 25-30% cost reduction	Limited DT security; High complexity; Integration challenges
4	A Systematic Literature Review of Supervised Machine Learning Techniques for Predictive Maintenance in Industry 4.0 (2025)	216 papers; Real-world (90.7%) vs Synthetic (9.3%); C-MAPSS; PHM datasets	Neural Networks dominant; Random Forest frequent (2024); 73.8% data not public; Accuracy 84-99%	Black-box DL; Limited explainability; Scarcity of labeled failure data
5	An edge-cloud IIoT framework for predictive maintenance in manufacturing systems (2025)	Real-time data; 3-phase motor; SMARTHome synthetic; Large manufacturing unit	IntelliPdM + MongoDB; Random Forest: 0.9901; Real: 93-95%; 25-30% cost reduction	Synthetic data dependency; Limited cybersecurity; Adaptability challenges

LITERATURE REVIEW: 21 PAPERS

6	An explainable artificial intelligence model for predictive maintenance and spare parts optimization (2023)	AI4I (synthetic, 14 features, 10K rows, 3.39% failure)	Random Forest + LIME: 0.982; Ada Boost: 0.9991; AUC 1.0	Synthetic dataset; Binary classification only; Spare parts framework incomplete
7	An explainable data-driven optimization framework for industrial predictive maintenance scheduling (2026)	C-MAPSS, IMS Bearings, SECOM, Lebanese Industrial (871 assets)	XGBoost/LightGBM + SHAP; AUROC 0.92-0.96; 22% downtime reduction; 15% cost reduction	SHAP associational only; Scalability concerns; No real-time validation
8	An explainable predictive maintenance strategy for multi-fault diagnosis of rotating machines using multi-sensor data fusion (2024)	Custom test setup (3-phase motor, 9 faults, 13.77 crore points)	Random Forest + LIME; Multi-sensor: 0.99; Single-sensor: 0.97; SVM: 0.64	Test setup not industrial; Limited 9 fault types; Real-time validation pending
9	Analyzing the impact of edge, fog and cloud computing on predictive maintenance in the Industrial Internet of Things (2025)	Bibliometric: 1,281 publications; Scopus database	PRISMA SLR; 35.82% growth; China (1760 TC), India (930 TC)	Scopus only; No experimental validation; Limited practical details
10	AssetHub AI assistant: LLM-based querying for industrial assets (2026)	2 AAS datasets (124 & 123 assets); 120 NL questions; MongoDB + PostgreSQL	GPT-4o-mini + RAG; Overall: 90.83%; Easy: 94.12%; Medium: 93.75%; Hard: 83.33%	Proprietary LLM reliance; Metadata quality dependency; Limited scalability
11	Building of transformer-based RUL predictors supported by explainability techniques: Application on real industrial datasets (2026)	C-MAPSS, MetroPT (9 failures), Scania (23,550 vehicles, 1.12M obs)	Transformer-encoder + LSTM + LIME; MetroPT RMSE: 0.647-1.229; C-MAPSS RMSE: 12.52-15.05	SHAP technical issues; Limited MetroPT failures (only 9); Class imbalance

LITERATURE REVIEW: 21 PAPERS

12	Comparison of deep learning models for predictive maintenance in industrial manufacturing systems using sensor data (2025)	Rotary, Milling, Hydraulic systems (100K, 50K, 200K samples)	CNN-LSTM: Accuracy 96.1%, F1 95.2%; AUC 0.98; CNN: 95.2%; BiLSTM: 95.8%	Black-box DL; Limited interpretability; Generalization challenges
13	Decision-making systems improvement based on explainable artificial intelligence approaches for predictive maintenance (2025)	Wind farm (87,208 samples, 81 variables, 23 failures, 2 years)	AE-DEVS + PSO; True detection 100%; RMSE 1.05; Early detection up to 1 week	Wind domain only; No automatic time window selection; Data confidentiality
14	Deep Learning and Time Series Analysis for Internet of Things Device Predictive Maintenance (2024)	IoT sensor data (conceptual framework paper)	ARIMA + RNN/LSTM; Framework benefits only (qualitative)	No quantitative metrics; Conceptual only; Interpretability issues
15	End-to-End Architecture for Real-Time IoT Analytics and Predictive Maintenance Using Stream Processing and ML Pipelines (2025)	IoT sensor data (MARWAN HPC infrastructure)	Kafka + Spark + InfluxDB + MLflow; Latency: 50-70ms; Spark: 1.15-1.18s/batch	No accuracy metrics reported; Experimental only; Limited real-world validation
16	Explainable time series features for hard disk drive failure prediction (2025)	Backblaze SMART (18,802 drives, 471 failures, 23 attributes)	XGBoost + SHAP; FDR: 74.7%; FAR: 0.73%; Cost: 0.043 vs 0.100	Static model; No incremental learning; SHAP computational complexity
17	Explainable Predictive Maintenance: A Survey of Current Methods, Challenges and Opportunities (2024)	102 articles; Bearings, CMAPSS, Backblaze, Wind, Trains, Batteries	SHAP (25+), LIME (15+), LRP (8+); 70% XAI dominant; Anomaly: 45, Fault: 28, Prognosis: 21	No standardized metrics; Lack of human-centered evaluations; No unified framework

LITERATURE REVIEW: 21 PAPERS

18	Optimizing Predictive Maintenance in Industrial IoT Cloud Using Dragonfly Algorithm (2025)	AI4I (10K samples, 14 features, 5 failure modes)	BR-MDPO (Dragonfly); Accuracy 0.98; Precision 0.971; Recall 0.975; 68% error reduction	Linear regression assumption; Limited to Chinese enterprises; No non-linear dynamics
19	Predictive Maintenance in Industrial IoT Using Machine Learning Approach (2024)	Industrial IoT sensor data (conceptual framework paper)	ML overview; Jupyter implementation; Qualitative benefits only	No quantitative metrics; Conceptual only; No algorithm comparison
20	Reliability-centered approach to artificial intelligence-driven predictive maintenance for industrial Internet of Things (2026)	CSMAR (67 Chinese enterprises; 1,895 applications; 3,120 variables)	OLS regression; Adjusted R^2 0.78; 28% downtime reduction; 17% cost reduction	Linear regression limitation; Chinese context only; No cross-industry validation
21	Transitioning from Reactive to Predictive Maintenance in Mechanical Systems (2020)	Literature review; Developing economies context (Nigeria, Africa)	Conceptual framework; 30% cost reduction; 45% downtime reduction identified	Conceptual only; Limited empirical validation; Regional focus only

Research Gap Identified

Existing systems focus on prediction accuracy. None combine MongoDB-native analytics + Explainable AI + Maintenance Prioritization + Natural Language Interface.

DATASET DESCRIPTION

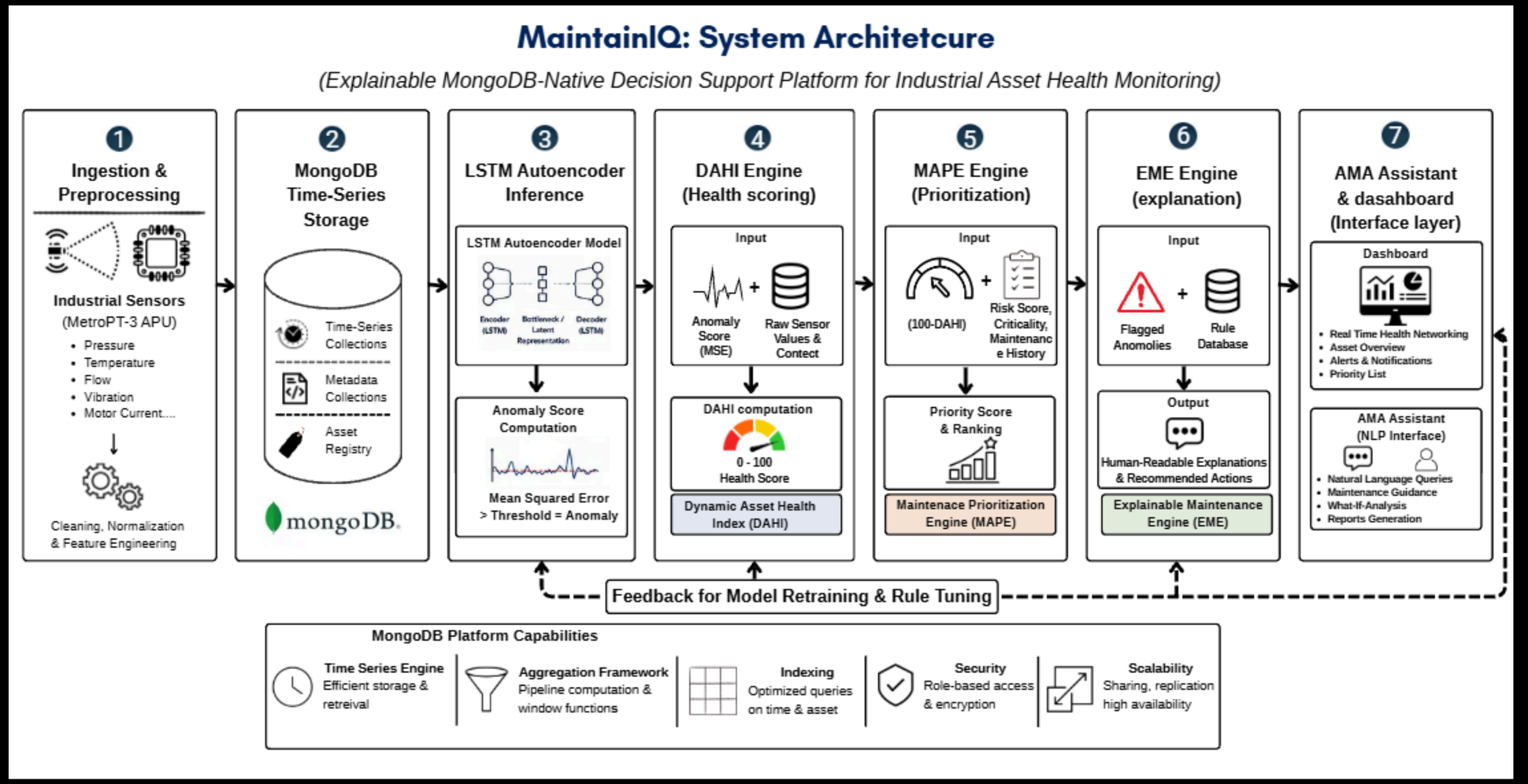
MetroPT-3 Dataset

Attribute	Value
Source	Metro Train APU, Portugal
Records	1,516,948
Features	15 (7 analog + 8 digital)
Failures	4 documented
Size	~218 MB

Features

No.	Feature	Type
1	TP2	Analog (bar)
2	TP3	Analog (bar)
3	H1	Analog (bar)
4	DV_pressure	Analog (bar)
5	Reservoirs	Analog (bar)
6	Motor_current	Analog (A)
7	Oil_temperature	Analog (°C)
8-15	COMP, DV_electric, TOWERS, MPG, LPS, Pressure_switch, Oil_level, Caudal_impulses	Digital

SYSTEM ARCHITECTURE DIAGRAM





REVIEW OF EXISTING MODELS

Method	Prediction	Explainability	Prioritization	NoSQL
Rule-Based	❌	✅	❌	❌
Isolation Forest	✅	❌	❌	❌
Autoencoder	✅	❌	❌	❌
LSTM	✅	❌	❌	❌
Traditional CMMS	❌	Limited	✅	❌
MaintainIQ	✅	✅	✅	✅

PROPOSED METHODOLOGY

Component	Description	Formula / Architecture	Key Output
1. LSTM Autoencoder	Unsupervised anomaly detection using reconstruction error	Input (60×10) → LSTM(64) → LSTM(32) → Bottleneck(16) → LSTM(32) → LSTM(64) → Output (Reconstructed)	Anomaly Score (MSE > 95th percentile = Anomaly)
2. DAHI Health Score	0-100 transparent health score per asset	$S_i = 100 \times (1 - \lvert x - \mu_i \rvert / \sigma_i)$ clipped to [0,100]; $DAHI = \Sigma(W_i \times S_i) / \Sigma(W_i)$	0-100 Health Score
3. MAPE Priority	Ranked maintenance queue based on urgency	$Priority = \alpha(100\text{-}DAHI) + \beta \times Risk + \gamma \times Criticality + \delta \times MaintenanceHistory$	Ranked Maintenance Queue
4. EME Explanations	Human-readable explanations with confidence	Rule-Based Engine (e.g., TP2↓ + Motor↑ → Air leak, Confidence 0.85)	Explanation + Action + Confidence
5. AMA Queries	Natural language to MongoDB query translation	5 Supported Query Types	Instant Answer to Natural Language

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Base Paper

- Alhammadi, M., Zaiter, I., & Jaradat, R. (2025). "An Autoencoder-Based Framework for Time-Series Anomaly Detection in Metro Train Air Production Units." IEEE GCAIoT 2025.

- **Dataset**
- Veloso, B., Ribeiro, R. P., Gama, J., et al. (2022). "The MetroPT dataset for predictive maintenance." Scientific Data, 9(764).
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Related Work

- Davari, N., Veloso, B., Ribeiro, R. P., Pereira, P. M., & Gama, J. (2021). "Predictive maintenance based on anomaly detection using deep learning for air production unit in the railway industry." IEEE DSAA 2021.
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